with $\mathbf{X} = (\boldsymbol{q}, \boldsymbol{p})^T$ and demonstrating that the required assumptions hold for (6.47). It is reasonable to consider periodic boundary conditions for the position variable, so $\mathcal{Q} = \mathcal{T}^{\mathrm{N_c}}$ where $\mathcal{T}^{\mathrm{N_c}}$ is an $\mathrm{N_c}$-dimensional torus. The use of periodic boundary conditions considerably simplifies the proofs, while also being more relevant to the practical realities of molecular dynamics simulation. The whole phase space is $\mathcal{D} = \mathcal{T}^{\mathrm{N_c}} \times \mathbb{R}^{\mathrm{N_c}}$.

We also consider the potential energy to be smooth (C^∞) with $U(\boldsymbol{q}) \to \infty$ as $|\boldsymbol{q}| \to \infty$. As we have a smooth potential on a bounded (position) domain, we have

$$1 < U_{\min} \leq U(\boldsymbol{q}) \leq U_{\max} < \infty, \quad \forall \boldsymbol{q} \in \mathcal{Q},$$

for some constants $U_{\min}$ and $U_{\max}$.

For the Lyapunov condition (Assumption 2), we choose the Lyapunov function

$$\phi(\boldsymbol{q}, \boldsymbol{p}) = H^l(\boldsymbol{q}, \boldsymbol{p}) = \left(\frac{1}{2} \|\boldsymbol{p}\|^2 + U(\boldsymbol{q}) \right)^l,$$

for a fixed positive integer l. As the potential energy is bounded below by 1, it is clear that $\phi(\boldsymbol{q}, \boldsymbol{p}) \to \infty$ as $|(\boldsymbol{q}, \boldsymbol{p})| \to \infty$. Computing the generator of Langevin dynamics as

$$\mathcal{L}_{\mathrm{LD}} f = \boldsymbol{p} \cdot \nabla_q f - \nabla U(\boldsymbol{q}) \cdot \nabla_p f - \gamma \boldsymbol{p} \cdot \nabla_p f + \gamma \mathrm{k_B T} \Delta_p \phi,$$

we can see that

$$\boldsymbol{p} \cdot \nabla_q \phi - \nabla U(\boldsymbol{q}) \cdot \nabla_p \phi = \mathcal{L}_H \phi = 0,$$

where $\mathcal{L}_H$ is the Lie derivative of our original Hamiltonian system (which preserves all functions of the Hamiltonian). Hence

$$\mathcal{L}_{\mathrm{LD}} \phi = -\gamma \boldsymbol{p} \cdot \nabla_p \phi + \gamma \mathrm{k_B T} \Delta_p \phi.$$

Using $\|\boldsymbol{p}\|^2 < 2H$, we compute derivatives to find

$$\begin{aligned} \boldsymbol{p} \cdot \nabla_p \phi &= l H^{l-1} \|\boldsymbol{p}\|^2 = 2l\phi - 2l H^{l-1} U(\boldsymbol{q}), \\ \Delta_p \phi &= l(l-1) H^{l-2} \|\boldsymbol{p}\|^2 + \mathrm{N_c} l H^{l-1} \leq l(l + \mathrm{N_c} - 1) H^{l-1}. \end{aligned}$$

As H^l grows faster than H^{l-1}, for any fixed l we are able to find positive constants c and d such that

$$l(l + \mathrm{N_c} - 1) H^{l-1} \leq \frac{l}{2} H^l + c, \quad 2l H^{l-1} U(\boldsymbol{q}) < 2l H^{l-1} U_{\max} \leq \frac{l}{2} H^l + d.$$